

Step 2: Mathematical Model Formulation

This technical report establishes the comprehensive, consolidated academic mathematical formulation for the Time-Dependent Heterogeneous Vehicle Routing Problem with Time Windows (TDHVRPTW) representing our 25-node regional distribution network. This formulation serves as the formal logic framework required to guide computational solver engines and systematically verify fleet and network routing restrictions.

1. Model Scope & Hard Parameters

The operational delivery network behaves under a strict set of deterministic parameters derived directly from supply chain data collection and planning phases:

Network Nodes (V): The spatial distribution layout is modeled as a complete directed graph where $V = (0, 1, 2, \dots, 24)$ containing 25 total system vertices. Node 0 represents the primary Central Dr. Oetker manufacturing plant and depot facility situated at Lutterstraße 14, Bielefeld. Nodes 1 through 24 represent pre-mapped regional Lidl retail locations within the Ost Westfalen-Lippe (OWL) economic catchment zone.

Spatial and Temporal Matrices (c_{ij} , t_{ij}): The precise driving distance c_{ij} (measured in kilometers) and corresponding baseline free-flow travel time t_{ij} (measured in minutes) between any two individual vertices i and j are extracted from physical road network datasets using Google Maps driving calculations.

Standardized Unit Demands (q_i): To ensure standard vehicle capacity constraints, customer order quantities are measured using homogeneous Euro pallets as the immutable loading unit. Store replenishment demands q_i are structured into three discrete tiers based on physical store footprints and localized catchment areas: Low-Demand (3 pallets), Medium-Demand (4 pallets), and High-Demand (5 pallets). The total network grid demand across all 24 retail vertices equals exactly 99 pallets.

Demand-Dependent Unloading Service Times (s_i): The dwell time required to complete offloading and paperwork at each store location i is variable and dynamic. It is computed using the operational linear function $s_i = 10 + 2q_i$, where 10 minutes represents a fixed administrative base buffer and 2 minutes are added per handled pallet. This yields a service duration of 16 minutes for low-demand, 18 minutes for medium-demand, and 20 minutes for high-demand store locations.

Hard Customer Delivery Windows ([ei, li]): A uniform hard time window constraint is enforced across the customer grid. Inbound deliveries must arrive at store locations between the opening bound of 08:00 AM (ei = 480 minutes past midnight) and the closing bound of 12:00 PM (li = 720 minutes past midnight).

Maximum Route Horizon (Tmax): To strictly maintain compliance with driver shift regulations and preserve a robust buffer against unexpected highway delays, no single vehicle route can exceed an effective duration of 405 minutes. This operational parameter represents a standard 480-minute commercial shift minus a mandatory 45-minute driver rest period and a 30-minute traffic delay margin

2. Fleet Heterogeneity Definition

To maximize regional transportation resource efficiency, the delivery fleet is partitioned into two specific vehicle classes operating with disparate capacity ceilings (*K* = *K**heavy* U *K**medium*):		
Vehicle Category (Subset)	Pallet Capacity (Qk)	Operational Context & Fleet Alignment
Heavy Trucks (K heavy)	33 Pallets	Assigned to high-density store clusters and high-volume regional supply tracks.
Medium Trucks (K medium)	12 Pallets	Utilized for restrictive urban centers, narrow lanes, and low-volume local loops.

3. Time-Interval Congestion Multipliers

The operational scheduling timeline assumes all vehicles depart from the central depot at 07:00 AM (defined as minute 0 in the model clock tracking loop) . To replicate real-world commuter and freight congestion patterns within the Bielefeld and wider OWL road networks , travel times are dynamically scaled via a time-dependent step-function multiplier, $\mu(*t*)$, determined by the exact elapsed time instance a vehicle initiates travel from an origin vertex: Time Interval (Clock) Model Timeline (Minutes from 07:00) Congestion Factor (μ) Operational Justification

07:00 AM - 09:00 AM	0 to 120 minutes	1.3 (30% Temporal Delay)	Morning peak commuter rush hour across regional network arteries (A2/B61).
09:00 AM - 11:30 AM	120 to 270 minutes	1.0 (Free-Flow Baseline)	Standard mid-morning off-peak operational distribution window.

11:30 AM -12:00 PM 270 to 300 minutes 1.1 (10% Temporal Delay) Early lunch-hour regional commercial traffic surge.

4. Index of Mathematical Symbols

Σ (Sigma): Summation operator denoting the compounding addition of values across a defined set index.

$\forall$ (For All): Universal quantifier establishing that a mathematical constraint condition must apply uniquely to every element within the specified parent set.

$\cup$ (Union): Combines multiple distinct vehicle sets into a single unified group.

$\in$ (Element of): Set membership indicator linking an active individual index parameter to its parent set matrix.

x_{ijk} (Binary Decision Variable): Primary binary tracking variable. Equal to 1 if vehicle k travels directly from location i to location j ; equal to 0 otherwise.

T_{ik} (Continuous Scheduling Variable): Specifies the exact calculated time instance (expressed as minutes past midnight) when vehicle k arrives at node i .

5. The Optimization Equations

A. The Objective Function The mathematical goal minimizes the cumulative operational distance driven across all active vehicles within the unified fleet network:

Minimize $Z = \sum_{k \in K} \sum_{i=0}^{24} \sum_{j=0}^{24} c_{ij} \cdot x_{ijk}$

Justification: This objective aggregates every selected routing arc. Minimizing Z forces the optimizer to identify the absolute global distance minimum across the 25x25 spatial grid while prioritizing compact customer groupings.

B. Network Flow & Structural Operational Constraints

1. Node Visit Exactness

$\sum_{k \in K} \sum_{j=0}^{24} x_{ijk} = 1 \quad \forall i \in \{1, 2, \dots, 24\}$

This constraint mandates that each regional Lidl store vertex is visited exactly once by a single vehicle during the active shift. It mathematically prohibits a node from being completely omitted or receiving duplicate deliveries.

2. Network Flow Conservation

$$\sum_{j=0}^{24} x_{ijk} - \sum_{j=0}^{24} x_{jik} = 0 \quad \forall i \in \{0, 1, \dots, 24\}, k \in K$$

Justification: This maintains logical routing continuity. If a vehicle arrives at a customer node, it must subsequently depart from that same node. Vehicles are blocked from spawning or terminating inside a customer location.

3. Depot Origin and Termination Limits

$$\sum_{j=1}^{24} x_{0jk} \leq 1 \quad \text{and} \quad \sum_{i=1}^{24} x_{i0k} \leq 1 \quad \forall k \in K$$

Justification: This enforces that every truck k launches from the central Dr. Oetker manufacturing plant (Node 0) at most once and must terminate its delivery path by returning to Node 0. Unused vehicles are held at 0.

4. Heterogeneous Vehicle Capacity Bounds

$$\sum_{i=1}^{24} q_i (\sum_{j=0}^{24} x_{ijk}) \leq Q_k \quad \forall k \in \dots K \text{ where: } Q_k = 33 \text{ if } k \in K \text{ heavy, and } Q_k = 12 \text{ if } k \in K \text{ medium}$$

Justification: This handles payload thresholds. It sums the pallet demands assigned to route k and verifies the volume against that specific truck type's physical maximum, restricting medium vehicles to 12 pallets and heavy units to 33 pallets.

5. Time-Dependent Timeline Propagation & Subtour Elimination

$$T_{ik} + s_i + [\mu (T_{ik} + s_i) \cdot t_{ij}] - M(1 - x_{ijk}) \leq T_{jk} \quad \forall i, j \in \{0, 1, \dots, 24\}, \forall k \in K$$

where the congestion step-function is modeled as:

$$\mu(t) = 1.3 \text{ if } 0 \leq t < 120 \text{ (07:00 AM - 09:00 AM)}$$

$$\mu(t) = 1.0 \text{ if } 120 \leq t < 270 \text{ (09:00 AM - 11:30 AM)}$$

$$\mu(t) = 1.1 \text{ if } 270 \leq t \leq 300 \text{ (11:30 AM - 12:00 PM)}$$

Justification: Formulated via the Big-M method ($M = 99,999$). When arc translation occurs ($x_{ijk} = 1$), the large scalar cancels out, advancing arrival time T_{jk} by accounting for travel time scaled by the active congestion coefficient μ . This function also mathematically breaks invalid subtour loops detached from the depot.

6. Customer Operating Windows

$$480 \leq T_{ik} \leq 720 \quad \forall i \in \{1, 2, \dots, 24\}, k \in K$$

Justification: Binds vehicle arrival variables to customer operating constraints, forcing all store arrival times strictly within the 480-minute (08:00 AM) to 720-minute (12:00 PM) window.

7. Shift Duration Limits

$$T_{0k}^{\text{return}} - T_{0k}^{\text{departure}} \leq 405 \quad \forall k \in K$$

Justification: Limits cumulative shift duration to a maximum of 405 effective minutes, checking the total elapsed time from original depot departure to final inbound return.